Language Concreteness

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What is language concreteness?

Brysbaert et al. (2014) argue that concreteness measures the degree of a concept denoted by a word that refers to a perceptible entity. In contrast to abstract words, concrete words are psycholinguistically easier for people to remember. In psychological research, concreteness has been an important and popular variable due to the availability of norms of words.

Previous research related to concreteness measurements includes the MRC database (Coltheart, 1981), English Lexicon Project (Balota et al., 2007), and the British Lexicon Project (Keuleers, Lacey, Rastle, & Brysbaert, 2012). But they are limited in word samples, word choices, and measurements in the concreteness ratings. For example, in measuring the adjective, five senses including touch, hearing, seeing, smelling, and tasting are not equivalently presented in the concreteness rating. So to improve the existing limitations from previous concreteness measuring approaches and scales, Brysbaert et al. collected new ratings for larger numbers of stimuli and developed a more standard word list for future studies. In the experiment, participants used a **5-point rating scale from abstract**, **1**, (language-based) to **concreteness**, **5**, (experience-based) to measure their feelings of words. In total, they measured about 40 thousand (37,058 words and 2,896 two-word expressions) generally known English word lemmas.

What do concreteness results tell us?

This file is the result of Brysbaert et al. research, which includes about 40 thousand words with their concreteness.

	A	B	C	D	E	F	G	H	I	J	K	L
1	Word	Bigram	Conc.M	Conc.SD	Unknown	Total	Percent_	SUBTLEX	Dom_Pos			
2	roadsweeper	0	4.85	0.37	1	27	0.96	0	0			
3	traindriver	0	4.54	0.71	3	29	0.9	0	0			
4	tush	0	4.45	1.01	3	25	0.88	66	0			
5	hairdress	0	3.93	1.28	0	29	1	1	0			
6	pharmaceutics	0	3.77	1.41	4	26	0.85	0	0			
7	hoover	0	3.76	1.23	4	29	0.86	162	0			
8	shopkeeping	0	3.18	1.19	1	29	0.97	0	0			
9	pushiness	0	2.48	1.24	1	30	0.97	0	0			
10	underdevelop	0	2.37	1.4	0	30	1	0	0			
11	tirelessness	0	2.28	1.28	1	30	0.97	0	0			
12	oldfashioned	0	2.26	1.02	0	27	1	0	0			
13	wellmannered	0	2.25	1.14	2	30	0.93	0	0			
14	dismissiveness	0	1.83	1	1	30	0.97	0	0			
15	spitefulness	0	1.8	0.76	0	25	1	0	0			
16	untruthfulness	0	1.73	0.92	4	30	0.87	0	0			
17	dispiritedness	0	1.56	0.71	3	28	0.89	0	0			
18	sled	0	5	0	0	28	1	149	Adjective			
19	plunger	0	4.96	0.2	0	26	1	48	Adjective			
20	human	0	4.93	0.26	0	28	1	6363	Adjective			
21	waterbed	0	4.93	0.27	0	27	1	27	Adjective			
22	cymbal	0	4.92	0.28	1	25	0.96	13	Adjective			
23	ginger	0	4.92	0.27	0	26	1	327	Adjective			
24	bobsled	0	4.9	0.41	0	29	1	48	Adjective			
25	cardboard	0	4.9	0.41	0	29	1	138	Adjective			
26	olive	0	4.9	0.31	0	30	1	375	Adjective			
27	dogsled	0	4.89	0.32	0	27	1	2	Adjective			
28	rubber	0	4.86	0.74	0	29	1	714	Adjective			
29	soybean	0	4.82	0.77	0	28	1	25	Adjective			
30	tangerine	0	4.81	0.62	0	27	1	38	Adjective			
31	headrest	0	4.8	0.76	0	30	1	6	Adjective			
32	eucalyptus	0	4.77	0.57	0	30	1	25	Adjective			
33	saltwater	0	4.77	0.82	0	30	1	31	Adjective			

NLP Suite → lib → concretenessLib → Concreteness_ratings_Brysbaert_et_al_BRM.csv

Word: the word

Bigram: whether it is a single word or a two-word expression

Conc.M: the mean concreteness rating

Conc.SD: the standard deviation of the concreteness ratings

Unknown: the number of participants indicating they did not know the word

Total: the total number of participants who rated the word

Percent: percentage participants who knew the word

SUBTLEX: the SUBTLEX-US frequency count (on a total of 51 million; Brysbaert & New, 2009)

Dom_Pos: the POS tagging (adjective, adverb, article, conjunction, determiner...)

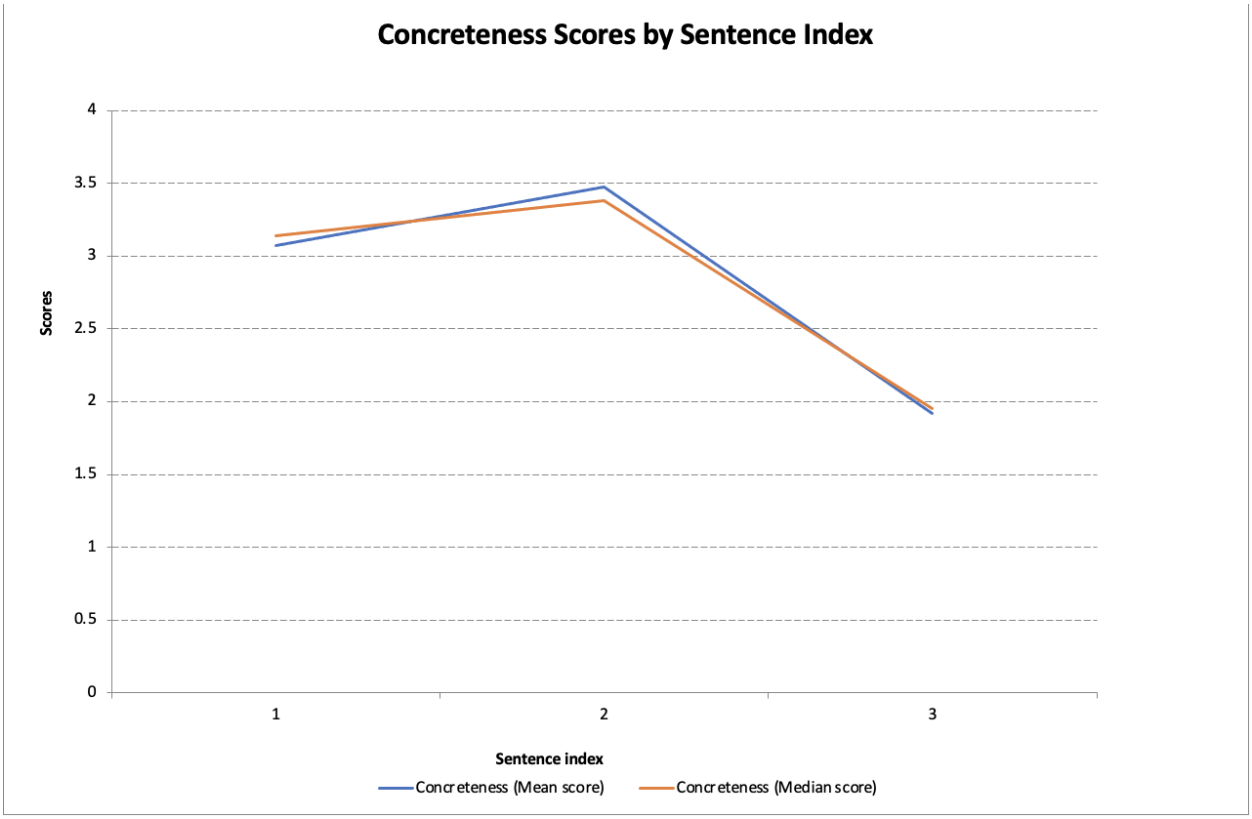
How to understand the output file

To calculate the concreteness for each sentence in the text file, the algorithm would count the number of words found in Concreteness_ratings_Brysbaert_et_al_BRM.csv (# Words Found, Percentage). Then, the mean and median concreteness scores are computed.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Document ID	Docum	Sentence ID	Sentence	Concreteness (Mean score)	Concreteness (Median score)	Standard Deviation	# Words Found	Percentage	Found Words	All Words						
2	1	/Users/	1	As an introduction to	3.075714286	3.14	1.055722241	14 out of 17	82.35%	(introduction, 3.0), (conundr	introduction, conundrum, complexity, ali, smith, bicameral, novel, consider, conversation, h						
3	1	/Users/	2	The mother is dead, a	3.478333333	3.38	0.778316559	6 out of 7	85.71%	(mother, 4.6), (dead, 4.07), (c	mother, dead, conversation, take, place, george, memory						
4	1	/Users/	3	But let's not get too c	1.92	1.955	0.47644517	4 out of 4	100.00%	(let, 2.26), (get, 2.38), (compl	let, get, complicate, yet						
5	1	/Users/	4	In George's memory,	N/A	N/A	N/A	0	0.00%	N/A	['george']						
6	1	/Users/	5	In George's memory,	2.806666667	3	0.370045042	3 out of 5	60.00%	(memory, 2.38), (last, 3.04), (	george, memory, last, may, italy						
7	1	/Users/	6	Mother, daughter and	3.263333333	3.3	1.117169041	15 out of 20	75.00%	(mother, 4.6), (daughter, 4.7	mother, daughter, george, brother, henry, travel, south, home, cambridge, england, see, ea						
8	1	/Users/	7	Mother and daughter	4.0275	4.335	0.967552755	4 out of 5	80.00%	(mother, 4.6), (daughter, 4.7	mother, daughter, talk, restoration, fresco						
9	1	/Users/	8	As Ms. Smith writes it: "But which came first?	3.3775	3.285	0.755088295	4 out of 4	100.00%	(smith, 3.81), (write, 4.22), (c	smith, write, come, first						
10	1	/Users/	9	Her mother says	3.59	3.59	1.428355698	2 out of 2	100.00%	(mother, 4.6), (say, 2.58)	mother, say						
11	1	/Users/	10	The chicken or the eg	4.885	4.885	0.120208153	2 out of 2	100.00%	(chicken, 4.8), (egg, 4.97)	chicken, egg						
12	1	/Users/	11	The picture underneat	4.2525	4.39	0.381870397	4 out of 4	100.00%	(picture, 4.52), (underneath,	picture, underneath, picture, surface						
13	1	/Users/	12	The picture below can	3.145	2.74	0.919909416	4 out of 5	80.00%	(picture, 4.52), (come, 2.72), (	picture, come, first, george, say						
14	1	/Users/	13	Because it was done f	2.61	2.61	0.212132034	2 out of 2	100.00%	(do, 2.46), (first, 2.76)	do, first						
15	1	/Users/	14	But the first thing we	3.4	3.19	0.656031842	10 out of 10	100.00%	(first, 2.76), (thing, 3.17), (see	first, thing, see, mother, say, time, thing, see, one, surface						
16	1	/Users/	15	So does that mean it	2.66	2.72	0.14	3 out of 3	100.00%	(mean, 2.5), (come, 2.72), (fir	mean, come, first						
17	1	/Users/	16	And does that mean t	2.901666667	2.75	0.973456042	6 out of 6	100.00%	(mean, 2.5), (picture, 4.52), (t	mean, picture, know, may, well, exist						
18	1	/Users/	17	This same question, a	2.626666667	2.37	1.044234169	9 out of 10	90.00%	(question, 3.36), (many, 2.37	question, many, others, arise, halve, novel, though, drastically, different, circumstance						
19	1	/Users/	18	This is not a book to t	3.01	3.175	1.115445284	12 out of 12	100.00%	(book, 4.9), (read, 3.56), (pas	book, read, passively, even, read, especially, since, take, figure, two, part, connect						
20	1	/Users/	19	To amp up her games	3.377857143	3.445	0.685881562	14 out of 16	87.50%	(amp, 3.59), (smith, 3.81), (pu	amp, gamesmanship, smith, publish, two, edition, sequence, halve, vary, copy, copy, georg						
21	1	/Users/	20	"How to Be Both," a fi	3.075454545	3.25	1.108010502	11 out of 12	91.67%	(finalist, 3.79), (year, 3.25), (n	finalist, year, man, booker, prize, accessible, yet, ordinary, george, story, come, first						
22	1	/Users/	21	It's knottier and more	N/A	N/A	N/A	0	0.00%	N/A	['knottier']						
23	1	/Users/	22	It's knottier and more	2.73	2.56	0.930161276	13 out of 18	72.22%	(challenge, 2.11), (begin, 2.5	knottier, challenge, begin, read, francescho, del, cossa, renaissance, painter, whose, first, na						
24	1	/Users/	23	And it's knottier and i	N/A	N/A	N/A	0	0.00%	N/A	['knottier']						
25	1	/Users/	24	And it's knottier and i	3.324666667	2.9	1.239768796	15 out of 17	88.24%	(challenge, 2.11), (wonder, 1.	knottier, challenge, wonder, artist, keep, conjure, girl, kind, magic, tablet, portrait, blonde, t						

Example from the New York Times Book Review Corpus (Spring 2021 Project)

The auto-generated output of the line chart shows the mean concreteness and the median concreteness score of all sentences given index 1, 2, and 3 in all text files. But we encourage you to look at the "Excel data" spreadsheet to manually create a line graph for more sentences.



Example from the New York Times Book Review Corpus (Spring 2021 Project)

References

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